

Application of Information & Communication Technologies

Lecture-12

Recap of Lecture 11

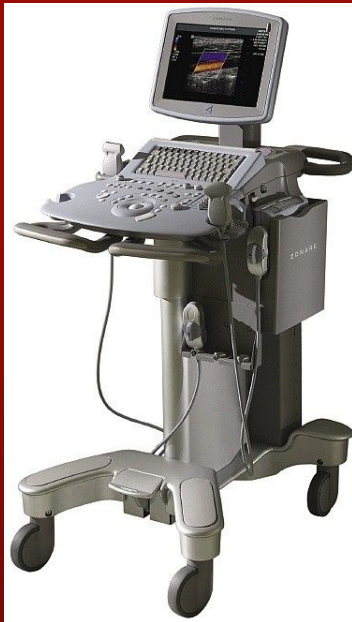
- ◆ System vs Application Software
 - Differences
 - Functions of OS
 - Types of OS

Overview of Lecture 12

- ◆ Other OS
- ◆ Utility Software
 - “Programs that help manage, maintain, and control computer resources.”

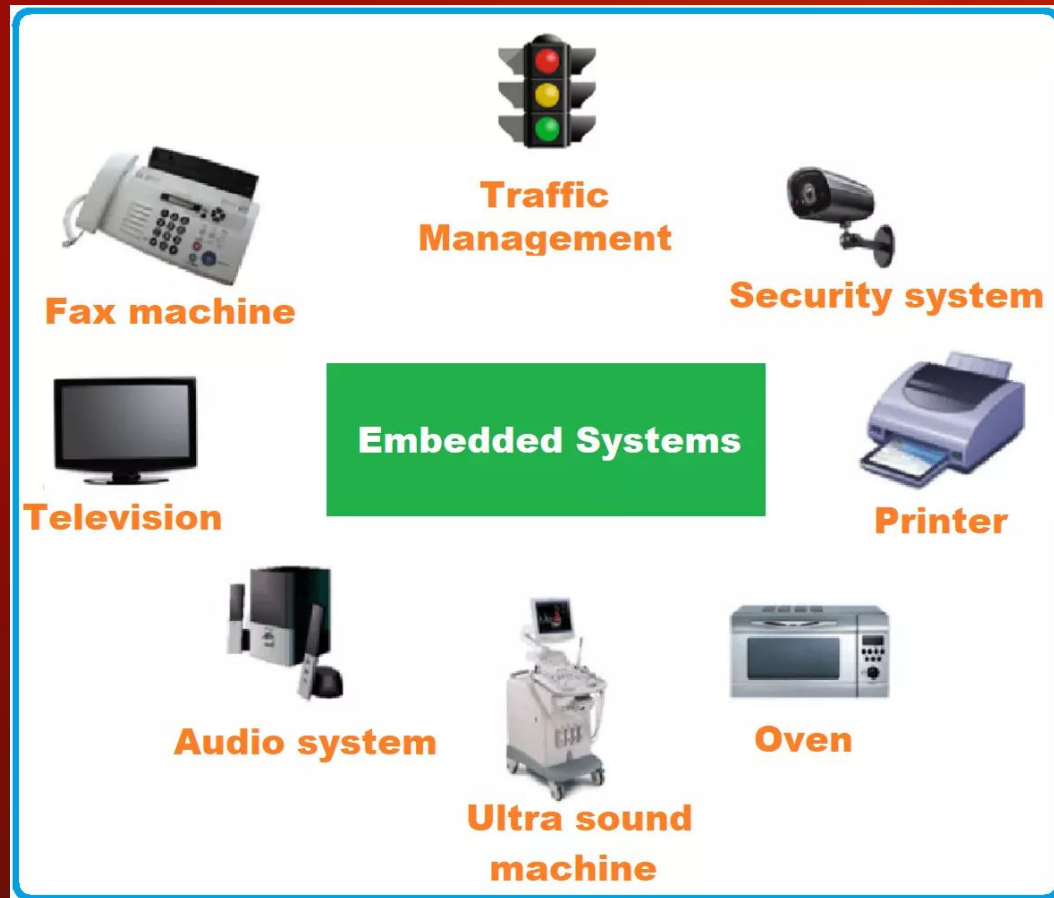
Other Operating System

- ◆ Real-Time OS (RTOS)
- ◆ Responds instantly to input.
- ◆ Used in medical devices, airplanes, robots.



Other Operating System

- ◆ Embedded OS Built into devices.
- ◆ Example: washing machine, ATM, Smart TV, Oven, Printer.



Other Operating System

- ◆ Mobile OS
- ◆ Designed for smartphones/tablets.
- ◆ Example: Android, iOS, windows phone.

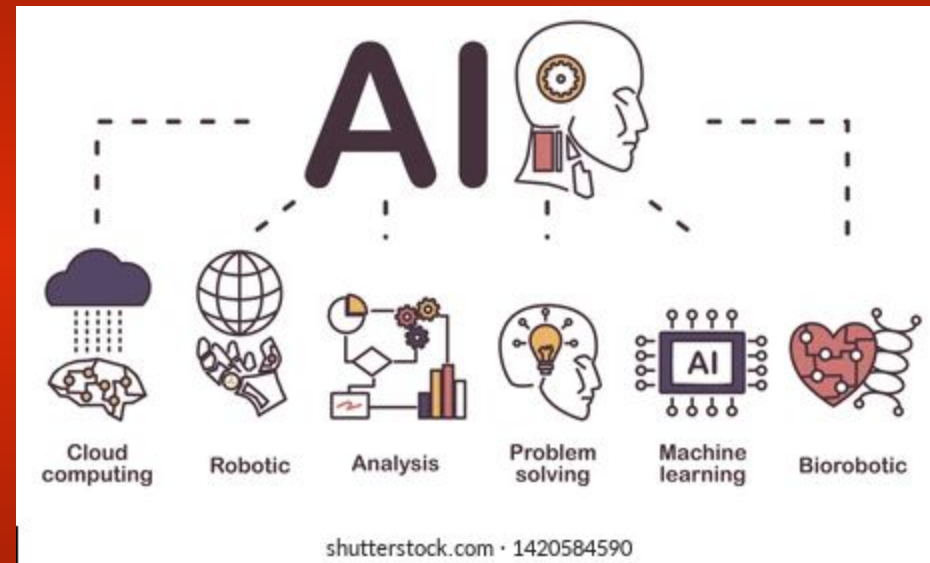


Summary Table

OS Type	Example	Use
Single-user	Windows	Personal PC
Multi-user	UNIX	Server
Real-time	VxWorks	Robots, medical
Embedded	ATM OS	Devices
Mobile	Android	Phones

Future of OS

- ◆ Cloud-based operating systems.
- ◆ AI-powered assistants.
- ◆ Hybrid OS (works on PC + phone).

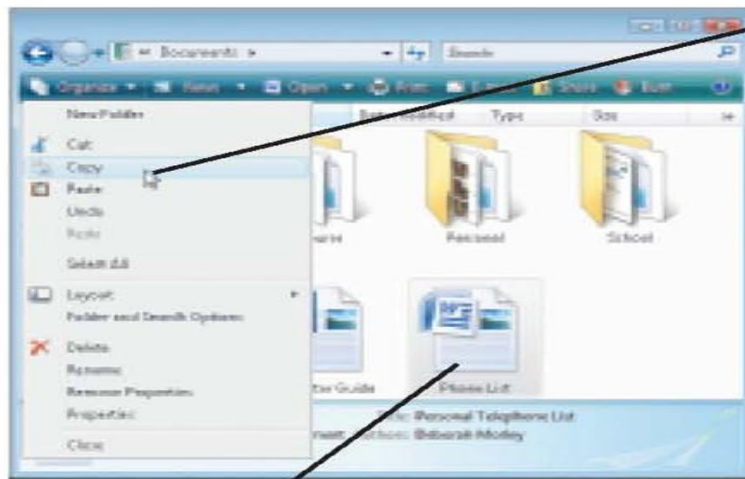


Utility software

- ◆ Utility software supports the operating system and performs a specific task.
- ◆ Helps in:
 - System performance improvement
 - File management
 - Security and backup
- ◆ Many utilities are built into operating systems (for finding files, viewing images, backing up files, etc.)

Using Windows Explorer

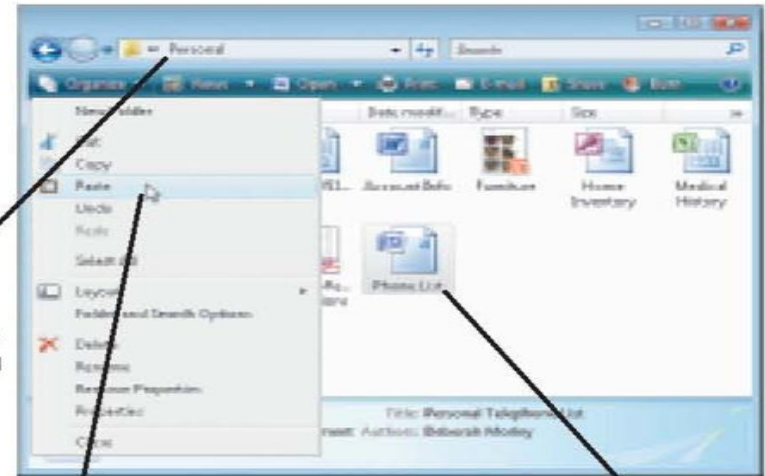
◆ Move the file



1. Navigate to the drive and folder containing the file you want to copy or move, and then select the file.

2. Click the Organize button and select *Copy* to copy the file to the Clipboard.

3. Navigate to the drive and folder where you want the file to go.



4. Click the Organize button and select *Paste* to copy the file to the current location.

5. The file is copied.

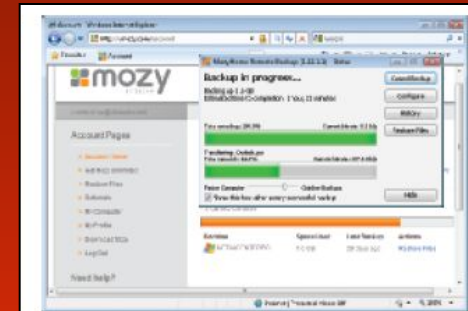


FIGURE 5-20

Using Windows Explorer to copy files.

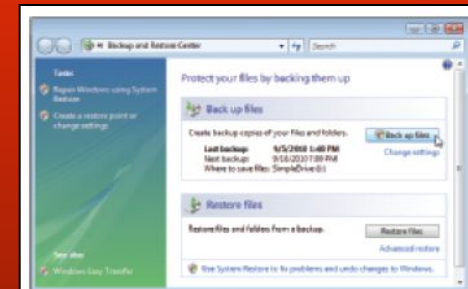
Backup and recovery utilities

- ◆ Make the backup and restoration process easier
 - **Backup:** Duplicate copy of data or other computer content
 - Good backup procedures are critical for businesses
 - Individuals should back up important documents, e-mail, photos, home video, etc.
 - Store backup data on a CD or DVD, second hard drive, flash memory drive, or upload to the Internet
 - Back up your entire computer once all programs have been installed, so your system can be restored to that configuration.



WEB-BASED BACKUP SERVICE

Allows you to back up files to a secure Web site.



WINDOWS BACKUP PROGRAM

Allows you to back up files to the desired backup medium manually or on a regular basis automatically.

FIGURE 5-22
Backup utilities.

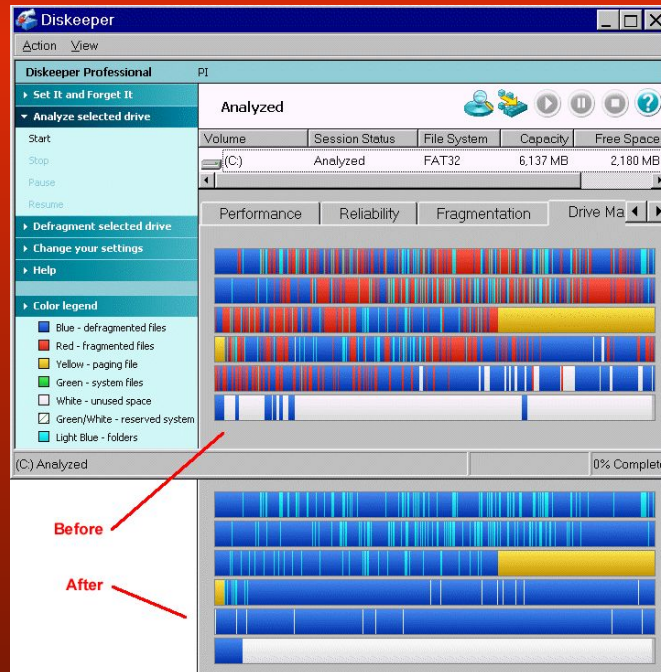
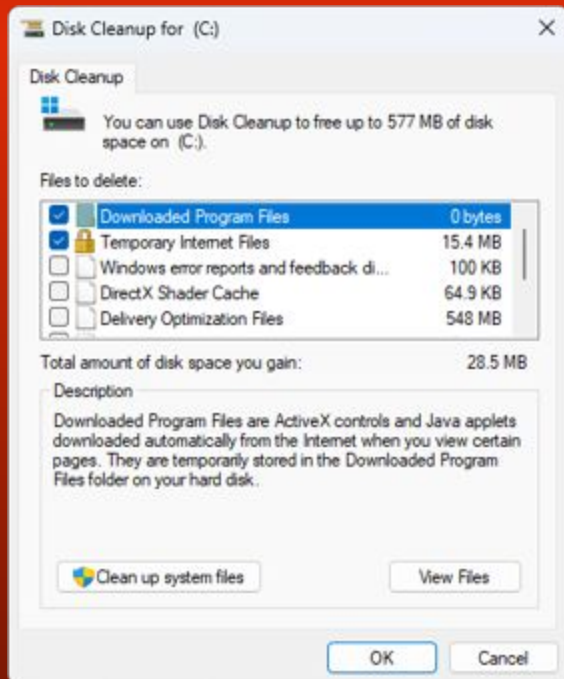
Antivirus and Security Utilities

- ◆ Protect against malware and viruses
- ◆ Examples: Avast, Kaspersky, Windows Defender



Disk Management Utilities

- ◆ Disk Cleanup: Removes temporary/unwanted files
- ◆ Disk Defragmenter: Reorganizes fragmented data for faster access
- ◆ Partition Managers: Create/resize/delete disk partitions



Disk Management Utilities

- ◆ Disk Defragmenter: Reorganizes fragmented data for faster access

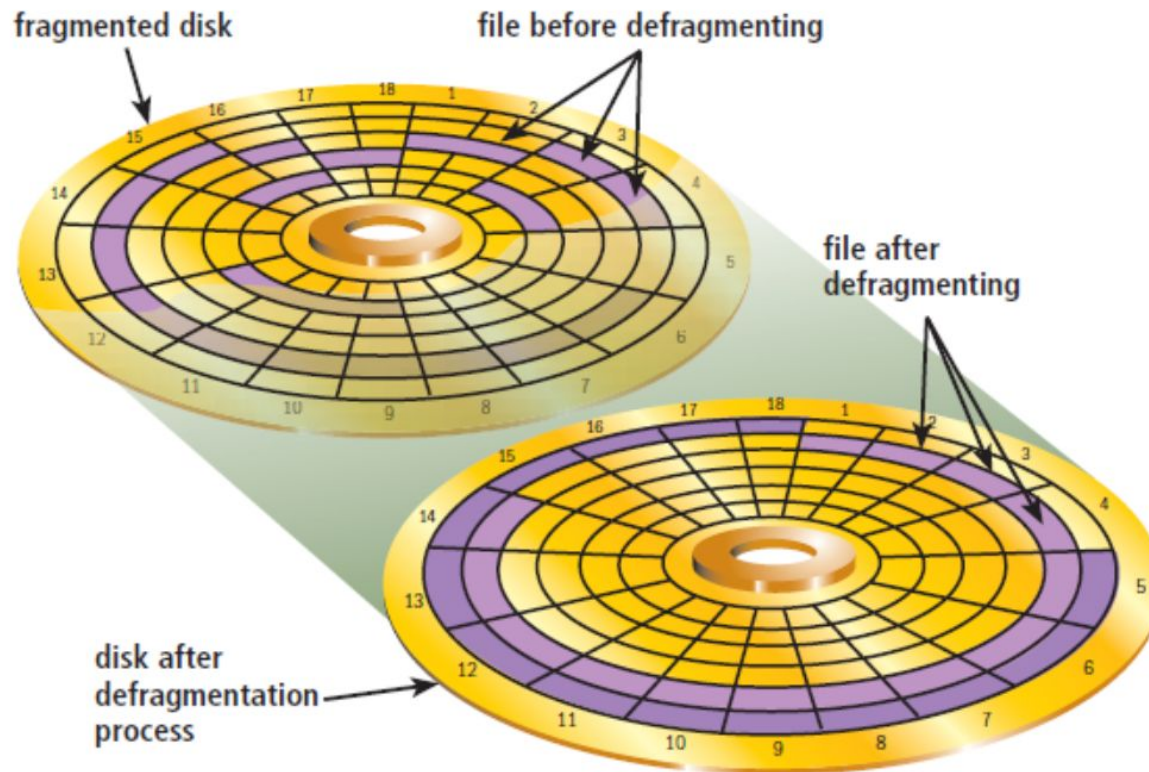
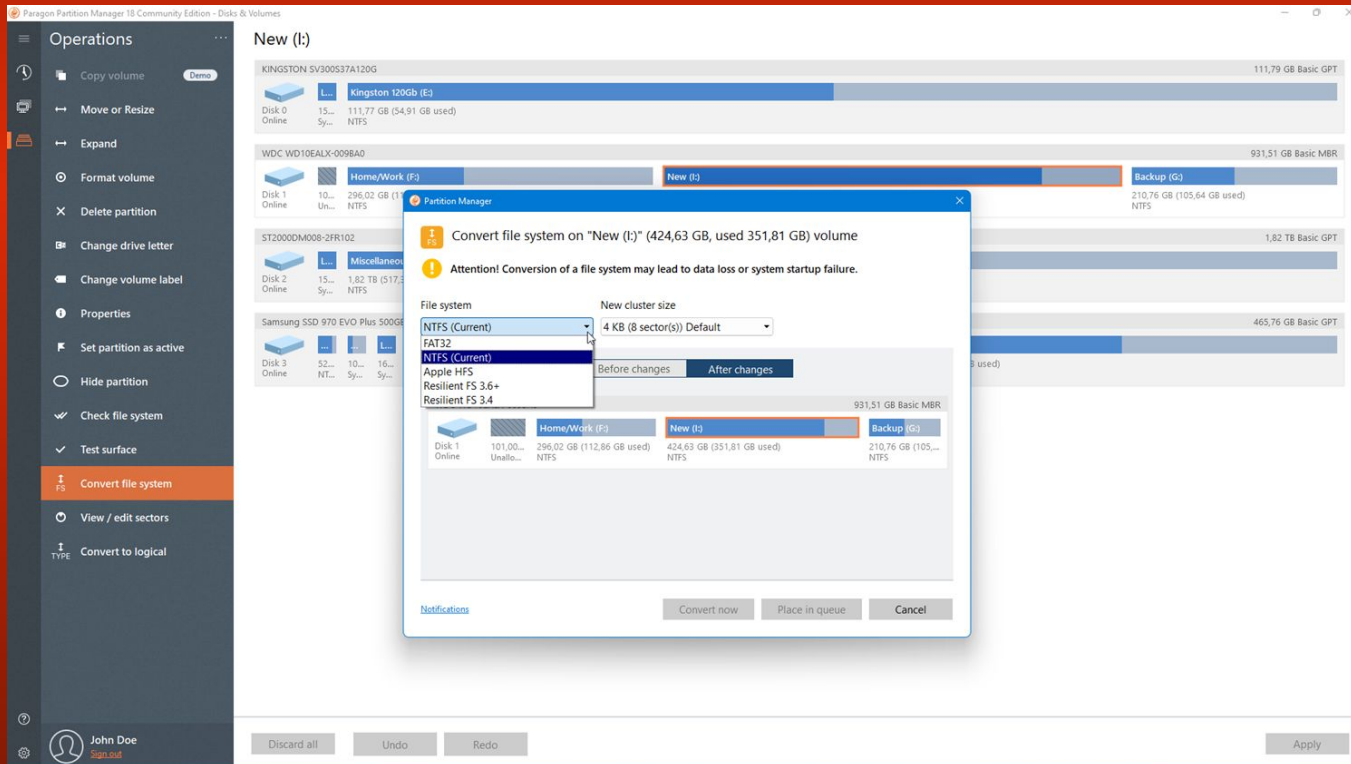


Figure 7-18 A fragmented disk has many files stored in noncontiguous sectors. Defragmenting reorganizes the files so that they are located in contiguous sectors, which speeds access time.

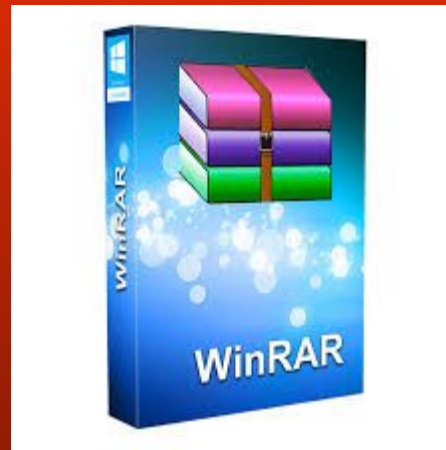
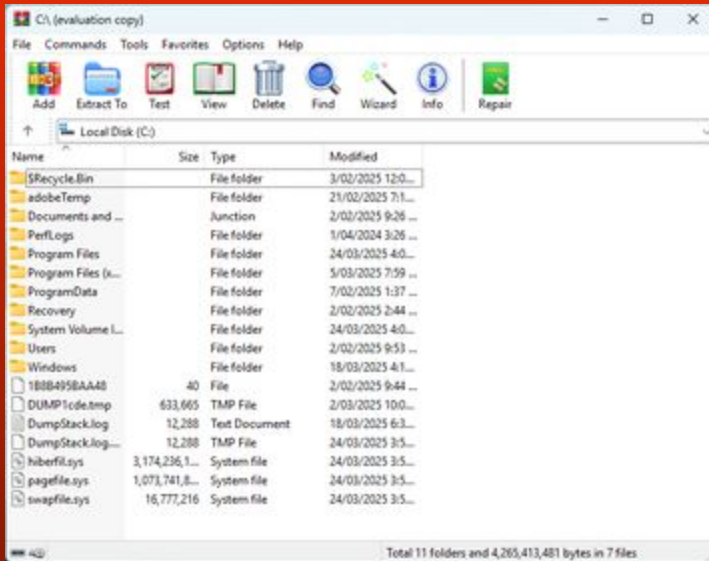
Disk Management Utilities

- ◆ Partition Managers: Create/resize/delete disk partitions



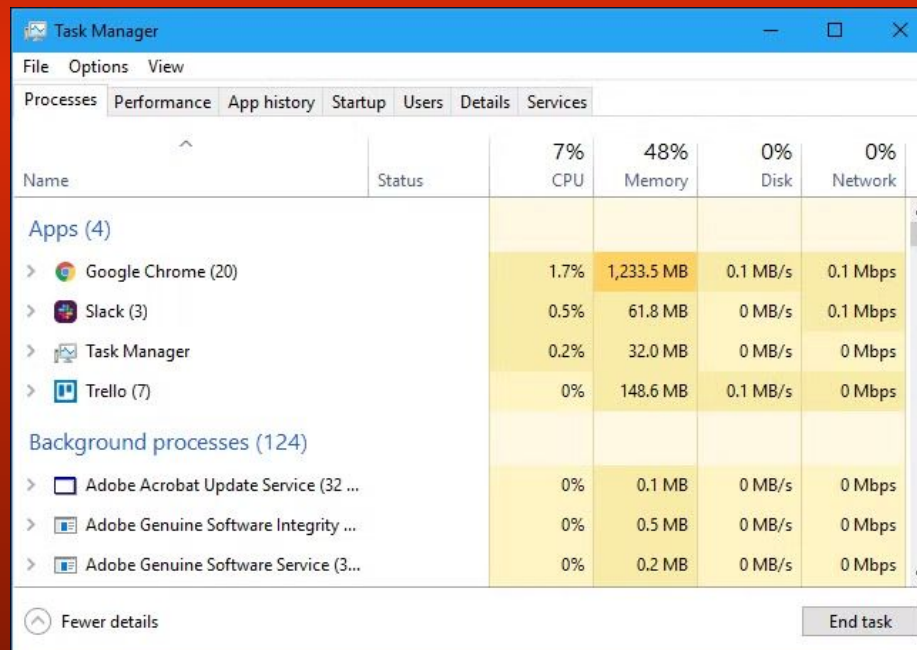
Compression Utilities

- ◆ Reduce file size for storage & transfer
- ◆ Examples: WinRAR, WinZip, 7-Zip
- ◆ Used for archiving and sending files faster



System Monitoring Utilities

- ◆ Task Manager, Resource Monitor
- ◆ CPU usage, memory usage, processes running
- ◆ Helps detect system performance issues

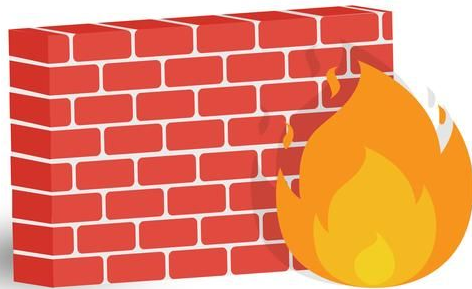


The screenshot shows the Windows Task Manager window with the 'Performance' tab selected. The window title is 'Task Manager'. The menu bar includes 'File', 'Options', and 'View'. The tabs at the top are 'Processes', 'Performance', 'App history', 'Startup', 'Users', 'Details', and 'Services'. The 'Performance' tab displays a summary of system resource usage: CPU at 7%, Memory at 48%, Disk at 0%, and Network at 0%. Below this summary is a table of running processes, categorized into 'Apps (4)' and 'Background processes (124)'. The table has columns for Name, Status, CPU, Memory, Disk, and Network. The 'Apps' section lists Google Chrome (20), Slack (3), Task Manager, and Trello (7). The 'Background processes' section lists Adobe Acrobat Update Service (32 ...), Adobe Genuine Software Integrity ..., and Adobe Genuine Software Service (3...). At the bottom of the window, there is a 'Fewer details' button and an 'End task' button.

Name	Status	7% CPU	48% Memory	0% Disk	0% Network
Apps (4)					
> Google Chrome (20)		1.7%	1,233.5 MB	0.1 MB/s	0.1 Mbps
> Slack (3)		0.5%	61.8 MB	0 MB/s	0.1 Mbps
> Task Manager		0.2%	32.0 MB	0 MB/s	0 Mbps
> Trello (7)		0%	148.6 MB	0.1 MB/s	0 Mbps
Background processes (124)					
> Adobe Acrobat Update Service (32 ...)		0%	0.1 MB	0 MB/s	0 Mbps
> Adobe Genuine Software Integrity ...		0%	0.5 MB	0 MB/s	0 Mbps
> Adobe Genuine Software Service (3...		0%	0.2 MB	0 MB/s	0 Mbps

Other Utilities

- ◆ Firewall Utilities (prevent unauthorized access)
- ◆ Uninstall Managers (completely remove programs)
- ◆ Data Recovery Tools (restore lost files)
- ◆ Cloud Backup Utilities (Google Drive, OneDrive)



FIREWALL



Summary

- ◆ Other OS
- ◆ Utility software = “Helpers of the Operating System”
- ◆ Improve security, performance, and data safety
- ◆ Examples: Antivirus, Disk Cleanup, Backup, Compression

Suggested Reading

- ◆ Ch-05, The System Unit: Processing and Memory ,
“Understanding Computers: Today and Tomorrow,
Comprehensive”, 15th Edition by Deborah Morley
& Charles S. Parker
- ◆ Ch-07, Discovering Computers Fundamentals-
Your Interactive Guide -- Gary B Shelly; Misty E
Vermaat; Jeffrey J Q